Multibody Syst Dyn (2012) 28:83–108
DOI 10.1007/s11044-011-9297-0

A multibody biomechanical model of the upper limb including the shoulder girdle

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Received: 11 May 2011 / Accepted: 16 December 2011 / Published online: 11 January 2012
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Abstract The aim of this work is to propose a robust musculoskeletal model of the upper limb to serve as the basis for the study of different types of shoulder pathologies, including the use of anatomical or reverse prostheses. The multibody biomechanical model is defined by seven rigid bodies constrained by the sternoclavicular, acromioclavicular, and glenohumeral joints, each modeled as a three d.o.f. spherical joint; the humeroulnar and radioulnar joints, each modeled as one d.o.f. hinge joint; and the scapulothoracic articulation, modeled by two holonomic constraints that allow the scapula to glide over the thorax. The muscle system includes 21 muscles described by 37 individual segments using the obstacle-set method. The muscle contraction dynamics is represented by the Hill-type muscle model, being the activation of each muscle unknown. The muscle force sharing is a redundant problem in which an optimization technique is applied to find the muscle activations, and the corresponding muscle forces, by minimizing an objective function that represents muscle energy consumption. The fulfillment of the equations of motion of the biomechanical model are enforced and the stability of the glenohumeral joint and the scapulothoracic articulation is also imposed, thus providing two sets of constraints for the optimal problem. The validation of the model is carried out by comparing the results from an acquired motion, the abduction of the arm, with available data in the literature and with EMG data.

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Keywords Multibody system dynamics · Inverse dynamic analysis · Musculoskeletal model · Upper limb · Shoulder girdle

1 Introduction

The upper limb is composed of an open chain mechanism that includes the shoulder, the elbow, and the wrist. The shoulder is the most complex and determinant mechanism of the limb, comprising four skeletal segments, the thorax, clavicle, scapula, and humerus, connected through three anatomical articulations: sternoclavicular (SC), acromioclavicular (AC) and glenohumeral (GH) joints, from medial to distal. In other words, it includes the shoulder girdle, clavicle and scapula, and the shoulder joint, scapula and humerus [1]. In the literature, a fourth articulation is often mentioned to describe the connection between the scapula and thorax. This articulation is not an actual anatomical joint though the scapula is constrained to slide over the thoracic cage [2]. The coordinated actions in all articulations give the shoulder mechanism the widest range of motion of all human body. However, such large flexibility comes at the cost of intrinsic joint stability being the shoulder joint, shown in detail in Fig. 1, the most frequently dislocated joint in the human body, despite the several mechanisms that provide additional stability to the glenoid fossa. Such stabilizing mechanisms include active muscle coordination, elastic ligament tension, labrum deformation, joint suction, adhesion/cohesion, articular version, proprioception, or negative internal joint pressure [3].

Although the upper limbs are needed for nearly all common activities of the upper body, the biomechanical models of the upper extremities are still at an early stage of development when compared to other regional investigations, such as gait and low back biomechanics [3, 4]. The first examples of biomechanical models for the upper limbs are reported at the end of the nineteenth century; physical models where the muscles of cadaver shoulder specimens were replaced with hemp threads allowed to determine moment arms and to qualitatively measure the contribution of the different muscles [4, 5]. Later on, the majority of studies of the shoulder focused on theoretical or computer models of varying complexity, though large-scale musculoskeletal models did not emerge until the 1980s [3, 6]. One of the most complete and notable dynamic models of the shoulder was presented in 1994 by Van der Helm [7]. Using a dynamic finite element method, bones were modeled as rigid bodies connected by three-dimensional hinge elements, and the scapulothoracic (ST) joint, which had been neglected until then, was originally modeled by constraining the scapula to remain in contact with an ellipsoid. Overall, the model included 16 muscles, each modeled by one to six force generating elements, 3 joints, 3 extracapsular ligaments, and the scapulothoracic gliding plane. More recently, other models have been presented [4, 8]. The model developed by Charlton and Johnson was comprised by 5 joints, which included the addition of the humeroulnar (HU) and radioulnar (RU) joints, 23 muscles, modeled by 93 muscle elements, and the scapulothoracic gliding plane. Even though the authors attempted to model the force produced by the costoclavicular and trapezoideus ligaments, the main goal and focus of the work was the validation of the developed model [4]. The model presented in [8] aimed at combining the capabilities of a musculoskeletal model with ergonomic analyses of the shoulder. The developed model was defined by 23 muscles, in a total of 38 muscle elements, 4 joints and the scapulothoracic gliding plane. The stability of the glenohumeral joint was the main novel aspect presented as it did not follow the previously published works where the projection

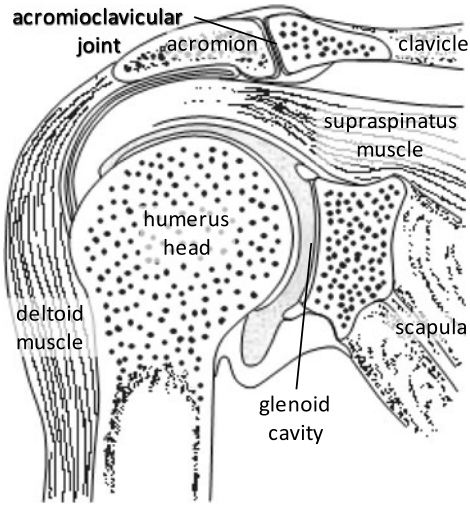


Fig. 1 Anatomical description of the shoulder complex

of the glenohumeral contact force was constrained to fall within an ellipse representing the glenoid cavity. In this work, an anisotropic, empirically-derived glenohumeral stability index that represents quantitatively the directional requirements of the glenohumeral interface for non-dislocation was applied. Regardless of the favorable aspects, the presented model was significantly limited in the simulation of the muscle contraction dynamics as the relationships force–length and force–velocity were not considered. Although many shoulder models exist in the literature, each has its own advantages and limitations, and thus several aspects of the shoulder biomechanics modeling are still open for further developments.

Optimization procedures are commonly used in multibody biomechanical models, with particular emphasis in locomotion applications [9–11]. These procedures, allied to the capability of the multibody systems to represent the complexity of the biomechanical systems, allow for the planning of rather elaborate human motions such as the overarm throw [12]. The identification of the motion objectives, through mathematical quantification, continues to be a research topic, as for instance on locomotion [13] or in investigations on human speech [14].

In this paper, a musculoskeletal model of the upper limb is developed using multibody system dynamics. The motivation behind this work is to build a robust model which can be continuously improved to allow simulating different pathological conditions, particularly when the shoulder has been replaced by an anatomic or reverse prosthesis, which is the third most common joint replacement [15].

2 Biomechanical model

2.1 Skeletal model

The upper limb model proposed here is based on data published by Garner and Pandy [16]. However, the reference frames proposed in reference [16] are not in agreement with the later standardization of the upper limb data proposed by the Standardization and Termi-

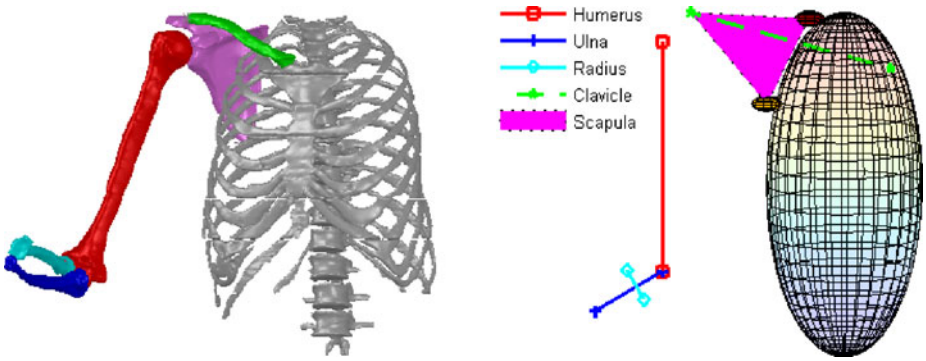


Fig. 2 (**a**) Geometrical description of the multibody system of the upper limb biomechanical model; (**b**) schematic representation of the upper limb

nology Committee of the International Society of Biomechanics (ISB). Using the available bony landmarks, new reference frames are defined in agreement with Wu et al. [17]. The reference frame of the scapula is based on the definition of Van der Helm [18] due to the lack of data of the original publications. Note that in order to improve the accuracy of the humerus segment coordinate system, the position of the forearm is changed from the standard position presented in [16]. The local coordinates of the bony landmarks described in the new references frames are reported in the Appendix.

The model proposed here includes the thorax, rib cage, clavicle, scapula, humerus, ulna, and radius modeled as rigid bodies, as depicted by Fig. 2. To overcome some deficiencies of the commonly used angular coordinates, the Euler parameters are adopted as orientational coordinates [19]. Therefore, each body is defined by 7 coordinates, leading to a generalized coordinates vector of 49 entries that represent the positions and orientations of all anatomical parts.

The thorax, defined as the global reference frame, and the rib cage, defined by an ellipsoid, is stationary. The SC, AC, and GH joints are each modeled as a 3 d.o.f. ball-and-socket joint. The ST joint is modeled using two holonomic constraints which ensure the scapula to glide over the rib cage, i.e., two points are defined at the surface of the scapula being their contact with the rib cage enforced at all time. The HU and RU joints are modeled as 1 d.o.f. hinge joints. Thus, by including the constraints resulting from the relationship between the Euler parameters, and all anatomical constraints aforementioned there is a total of 40 constraints in the upper limb model.

The body segment inertial parameters (BSIP) for the upper arm, humerus, and arm, ulna and radius, are estimated from predictive equations reported in the literature [20]. Regarding the scapula and the clavicle, there are no predictive equations so their BSIP are estimated from magnetic resonance imaging [21]. The masses and moments of inertia, defined in the principal axes of inertia, are shown in Table 1. For the scapula and the clavicle, the principal axes of inertia are considered to be aligned with the reference position, while for the remaining parts the orientation of these is taken from [20]. Tables 2 and 3 present the centers of mass (COM) and the direction cosines of the new reference frames for the position of the anatomical segments in the model.

Table 1 Masses and moments of inertia for the anatomical segments of the upper limb

Segment	Mass (kg)	I_{xx} (kg m^2)	I_{yy} (kg m^2)	I_{zz} (kg m^2)
Thorax	–	–	–	–
Clavicle	0.12784	0.00122800	0.00109900	0.0003900
Scapula	0.94898	0.01912900	0.03007200	0.0112420
Humerus	2.16000	0.01922569	0.00392116	0.0204860
Ulna	0.94883	0.00527229	0.00081371	0.0052723
Radius	0.58117	0.00322935	0.00049841	0.0032294

Table 2 Coordinates of the center of mass in the global reference frame for each body of the upper limb

Segment	x (m)	y (m)	z (m)
Thorax	0.000000	0.000000	0.000000
Rib cage	−0.058280	−0.168220	0.060660
Clavicle	−0.031891	0.045060	0.087043
Scapula	−0.101363	0.022263	0.133529
Humerus	−0.045643	−0.108735	0.174550
Ulna	0.068962	−0.271643	0.209029
Radius	0.066841	−0.264919	0.200488

Table 3 Direction cosines in the global reference frame for each body of the upper limb for the reference position represented in Fig. 2b

Segment	Axis	x (m)	y (m)	z (m)
Thorax	x	1.000000	0.000000	0.000000
	y	0.000000	1.000000	0.000000
	z	0.000000	0.000000	1.000000
Rib cage	x	0.191868	0.048710	0.980211
	y	0.972773	0.122858	−0.196517
	z	−0.129999	0.991228	−0.023812
Clavicle	x	0.945081	0.000000	0.326834
	y	0.150238	0.888086	−0.434432
	z	−0.290257	0.459677	0.839313
Scapula	x	0.700190	0.241287	0.671947
	y	0.030094	0.930349	−0.365435
	z	−0.713321	0.276096	0.644161
Humerus	x	0.972966	0.081387	0.216131
	y	−0.082770	0.996565	−0.002661
	z	−0.215605	−0.015299	0.976360
Ulna	x	0.000000	0.999191	−0.040203
	y	−0.976409	−0.008680	−0.215752
	z	−0.2159271	0.039254	0.975620
Radius	x	0.075004	−0.831538	−0.550380
	y	−0.996083	−0.088392	−0.002198
	z	−0.046821	0.548389	−0.834911

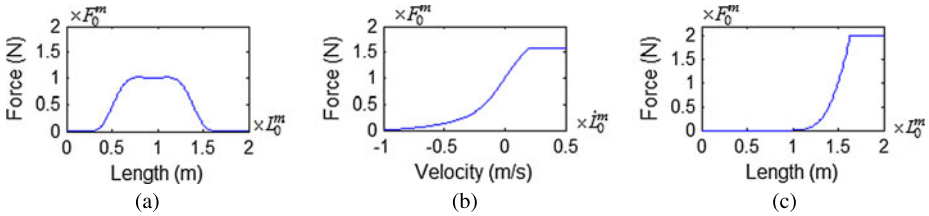


Fig. 3 Force–length and force–velocity relationships of the muscle model. (**a**) Force–length relationship of the contractile element; (**b**) force–velocity relationship representing a general concentric–eccentric contraction in which the force value is measured when the muscle length is equal to its rest length; (**c**) force–length relationship of the passive element

2.2 Musculoskeletal model

In this work, the muscle contraction dynamics is described by the Hill muscle model. The model is represented by an active contractile element (CE), simulating the active muscular action produced by the functional units of contraction, and a passive elastic element (PE), simulating the rigidity of the structures physically in parallel with the muscle fiber. Both of these elements are linked in series with a passive elastic element (SE), usually associated to cross-bridge stiffness, which can be neglected in coordination studies not involving short-tendon actuators [22, 23]. The muscle force is defined as the force produced by the CE element plus the contribution of the PE element:

$$\begin{cases} F^m = F^m_{\mathrm{CE}}\left(L^m(t), \dot{L}^m(t), a^m(t)\right) + F^m_{\mathrm{PE}}\left(L^m(t)\right) \\ F^m_{\mathrm{CE}} = \dfrac{F^m_L(t) F^m_{\dot{L}}(t)}{F^m_0} a^m(t) \end{cases} \tag{1}$$

The contractile properties of muscle tissue are controlled by its current length $L^m(t)$, rate of length change $\dot{L}^m(t)$, and muscle activation $a^m(t)$. The relationships muscle force–length $F^m_L(t)$ and force–velocity $F^m_{\dot{L}}(t)$, shown in Figs. 3a and 3b, are obtained from [22] and only depend on the kinematics of the muscle, i.e., its length and velocity of elongation. If the muscle kinematics and the muscle properties, in particular the optimal muscle-fiber length L^m_0, the tendon slack length L^m_s and the maximum isometric force F^m_0, are given, then the identification of the CE element contribution simply requires the evaluation of the level of muscle activation. The force produced by the passive element only depends on muscle length according to Fig. 3c.

The muscle geometry of 21 muscles, represented by 37 muscle segments, is implemented in the model and includes geometric features imposed by the obstacle-set method [24]. Additionally to usual via points, this method defines four types of objects, i.e., single sphere, single cylinder, double cylinder, and sphere-capped cylinder, to better describe the muscle path. In particular, when the muscle is constrained by bones and other muscles such basic geometries represent the anatomical geometries.

The upper limb musculoskeletal data is originally presented in [25], but since the body-fixed reference frames were changed, a new muscle data is now modified in a consistent manner and reported in the Appendix.

The muscles are introduced in the biomechanical model as point-to-point kinematic driver actuators. Each muscle is related to a constraint equation that forces the distance between two generic points of different rigid bodies to change according to a length-time

function [26]. For muscles defined by more than two points, their contribution to the biomechanical model is taken as a sum of several two-point muscle actuators that approximate their actual geometry.

2.3 Inverse dynamics

Inverse dynamics procedures are used in the solution of biomechanical problems when the motion is prescribed and only the forces responsible for such motion are unknown. The motion of the biomechanical model is usually acquired using video imaging or is defined by other criteria [26]. This is the case of the current study. The motion analyzed is acquired at the Laboratory of Biomechanics of Lisbon (LBL), at IST—TU Lisbon. The arm abduction is recorded for 7 seconds using four video cameras with sampling rates of 100 Hz. At the same time, and synchronously, electromyography (EMG) signals of 9 superficial muscles are recorded, namely the Deltoideus p. acromialis, Deltoideus p. clavicularis, Deltoideus p. scapularis, Biceps brachii, Serratus anterior, Trapezius p. descendenz, Trapezius p. transversus, Pectoralis Major, and Triceps brachii [27].

2.3.1 Kinematic data acquisition

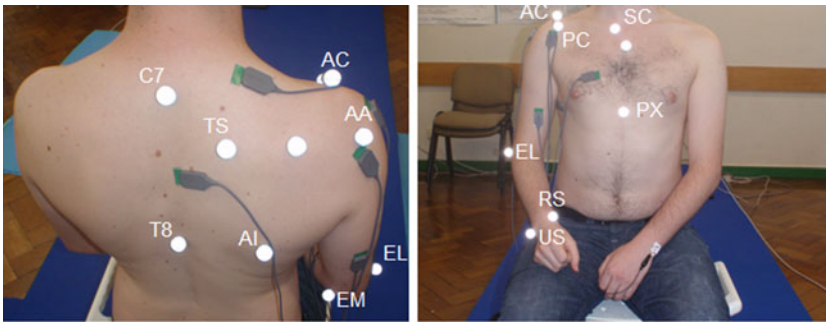
Based on the work by Van der Helm [18], the ISB proposed a standard procedure for the acquisition and description of the upper limb motion [17]. To record the three-dimensional motion of a rigid body, it is necessary to effectively describe its 6 d.o.f., i.e., the ideal situation would be having 3 anatomical landmarks, traceable during the complete motion period, for each bone. However, not all bones of the upper limb present 3 palpable bony landmarks and thus bony landmarks of other bones are needed for the definition of the segment coordinate systems, leading to a position-dependency. Wu et al. [17] recommend 15 bony landmarks, described in Table 4 and shown in Fig. 4, for the definition of the coordinate systems of the thorax, clavicle, scapula, humerus, ulna, and radius.

For the clavicle, only two bony landmarks are palpable so the axial rotation of the clavicle cannot be determined through noninvasive measurements. Taking as example [28], the axial rotation of the clavicle is estimated by minimizing the AC joint rotations. The scapula, even though it presents 3 palpable bony landmarks, does not allow their accurate tracking through the exclusive use of surface markers. During arm elevation, the scapula moves with respect to the skin causing a displacement between the marker and the corresponding anatomical landmark as large as 2 cm [29, 30]. The most accurate method involves the measurement of quasi-static positions, i.e., the landmarks are identified for successive static positions using a scapula locator. However, this method is too time consuming and the muscle activity may differ from the dynamic conditions [30, 31]. In this work, the scapula is tracked dynamically using a methodology recently proposed by Šenk and Chèze that show acceptable accuracy up to 120° of arm elevation, as seen in Fig. 4a [32]. It must be highlighted that the mean maximum humeral elevation appearing in a set of functional tasks is approximately 100° [33].

For the humerus, the glenohumeral joint center is needed to define its longitudinal axis, in particular. Since the GH joint center is not a palpable bony landmark, it has to be estimated by means of a regression analysis [34] or by other methods [35–37]. In this work, the algorithm presented in [30] is adopted due to its better accuracy [38].

Table 4 Anatomical landmarks of the thorax, clavicle, scapula, humerus, ulna and radius [17]

Bone	Bony landmark	Description
Thorax	C7	Processus Spinosus of the 7th cervical vertebra
	T8	Processus Spinosus of the 8th thoracic vertebra
	IJ	Deepest point of Incisura Jugularis
	PX	Processus Xiphoideus
Clavicle	SC	Most ventral point on the sternoclavicular joint
	AC	Most dorsal point on the acromioclavicular joint
Scapula	TS	Trigonum Spinae Scapulae
	AI	Angulus Inferior
	AA	Angulus Acromialis
	PC	Most ventral point of Processus Coracoideus
Humerus	GH	Glenohumeral rotation center
	EL	Most caudal point on lateral epicondyle
	EM	Most caudal point on medial epicondyle
Ulna	US	Most caudal-medial point on the ulnar styloid
Radius	RS	Most caudal-lateral point on the radial styloid



(a) (b)

Fig. 4 Position of the bony landmarks. (**a**) Posterior view; (**b**) anterior view

2.3.2 Equations of motion

Regardless of the method used to obtain the motion, this is only fully known if its characteristics are consistent with the kinematic constraints. Denoting by $\mathbf{q}$ the vector of the system coordinates, the kinematic constraints resulting from the kinematic joints and muscular actuators are represented as

$$\mathbf{\Phi} = \mathbf{\Phi}(\mathbf{q}) = \mathbf{0}. \tag{2}$$

The existence of kinematic constraints in the system implies the introduction of reactions forces between bodies. Following the detailed demonstration presented in [19], the reaction forces, denoted by $\mathbf{g}^{(c)}$, are expressed as

$$\mathbf{g}^{(c)} = \mathbf{\Phi}_{\mathbf{q}}^{T}\boldsymbol{\lambda} \tag{3}$$

where $\mathbf{\Phi_q}$ and $\boldsymbol{\lambda}$ represent, respectively, the Jacobian matrix of the constraints and the Lagrange multipliers vector related to the kinematic constraints.

For a constrained biomechanical model, the equations of motion are written as

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{\Phi}_\mathbf{q}^T \boldsymbol{\lambda} = \mathbf{g} \tag{4}$$

where $\mathbf{M}$ is the mass matrix of the system, which includes the mass and moment of inertia of all bodies; $\ddot{\mathbf{q}}$ is the acceleration vector; and $\mathbf{g}$ is the generalized external forces vector.

As mentioned before, the muscles are introduced in the system as kinematic drivers and not only as forces added to the force vector. The force produced by the muscle depends on the contribution of the contractile and passive element. Since motion is known for the complete analysis period, the passive element contribution, which only depends on muscle length, is also known and thus is treated as an externally applied force that acts directly onto the rigid bodies interconnected by the muscle. The contractile element contribution is taken into account as a reaction force, i.e., the Lagrange multipliers vector includes entries associated to the contractile element of each muscle. These Lagrange multipliers may represent muscle force or, if the muscle equation is multiplied by a proper scalar factor, muscle activation. Expressing the equations of motion in terms of activations has some advantages in the optimization problem since the design variables become bounded between 0 and 1 [23], thus enforcing them to have similar orders of magnitude.

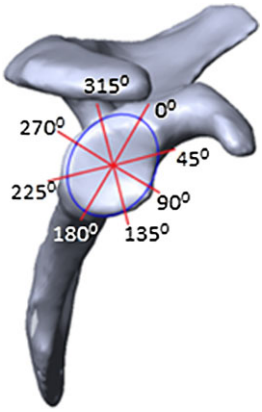
2.4 EMG analysis

The EMG signal depends upon several physiological, anatomical, and technical factors [39]. For instance, its amplitude is affected by muscle length, at the extreme positions of motion. Even though the muscle may be generating less force, the EMG amplitude may be higher due to the length of the muscle fibers [7]. As a result, an accurate quantitative relationship between the amount of force produced by the muscle and the EMG amplitude is more often than not elusive [39]. Still, the EMG signal is usually used to validate the results predicted from the inverse dynamic analysis. The qualitative relationship between the EMG amplitude and the muscle force allows one to define on/off patterns which can be compared to the model predictions [40].

The EMG signals recorded at the laboratory are rectified and its root-mean-square (rms) voltage calculated. To define the on/off patterns, the approach adopted includes the calculation of the standard deviation of the EMG baseline before any activity. Then a multiplication factor of this range is defined as the threshold, i.e., above this threshold the muscle is defined to be active and below it to be nonactive.

2.5 Muscle force sharing problem

The human musculoskeletal system is characterized by redundant muscles, i.e., different muscles may guide the same degree-of-freedom [41]. Mathematically, this means that the number of unknowns, the muscle activations, is larger than the number of degrees of freedom, which is a measure of the actual mobility of the system. The muscle force sharing problem has an infinite set of solutions, i.e., there is an infinite number of combinations of the muscle activations that lead to the same motion. The aim of the optimization procedure is to find the solution of the muscle force sharing problem that minimizes an objective function while subjected to a certain number of restrictions or constraints. Ideally, the objective



Direction (°)	Shear/compression ratio (%)
0	51
45	33
90	29
135	40
180	56
225	43
270	30
315	35

Fig. 5 Directional shoulder dislocation force threshold ratios. Ratios indicate the directional shear to compressive force joint dislocation tolerance [8]

function has some biological relevance, i.e., the goal is to simulate as much as possible the criteria adopted by the central nervous system when deciding which muscles to recruit and their levels of activation. Note though that the appropriate criterion to use depends on several aspects such as the type of motion under analysis, the objectives to achieve, or the presence of any type of pathology [26].

The most used optimization procedure, deemed as static optimization, considers the muscles as instantaneously available actuators whose force only depend on the current activation levels so that the optimization can be performed independently at each instant of time [42]. If the motion is discretized into $n-1$ constant time intervals, then n different optimization problems can be set. The procedure presented here, for the inverse dynamic analysis, attempts to minimize all motion at the same time rather than at each instant of time independently. Considering the analysis period to be discretized into n time steps, the design variables are the muscle activations for all m muscles at all n time steps. The $mn \times 1$ vector $\mathbf{a}$, called vector of global muscle activation, is arranged as

$$\mathbf{a} = \left[a_1^1, \ldots, a_j^1, \ldots, a_n^1, \ldots, a_1^i, \ldots, a_j^i, \ldots, a_n^i, \ldots, a_1^m, \ldots, a_j^m, \ldots, a_n^m\right] \tag{5}$$

where a_j^i is the activation of the ith muscle for time t_j.

Among the several objective functions with biological relevance available in the literature, it is adopted here the one proposed by Praagman et al. [43], which is an energy-related cost function. This function, based on the two major energy-consuming processes in the muscle, leads to better results than the most commonly used muscle stress cost function [43]. The problem is described as minimizing muscle energy consumption while satisfying the equations of motion of the biomechanical system and the muscle activations side constraints, which are null for the no-activation state, and unitary for the fully activated state. At the same time, the joints stability must also be assured. The glenohumeral joint is stabilized by applying directional empirically-derived stability ratios, shown in Fig. 5, to limit the amount of joint shear force relative to the compressive force [8] while the scapulothoracic joint is stabilized by assuring that the reaction forces are compressive [44].

Mathematically, the muscle force sharing problem is formulated as

$$\begin{aligned}
\min \quad & J(t,\mathbf{a}) = \sum_{t=1}^{n}\sum_{i=1}^{m} c_1 \frac{V^i F^i_{\mathrm{CE}}}{PCSA^i} + c_2 V^i a^i_t + c_3 {a^i_t}^2 \\
\text{subject to:} \quad & \begin{cases}
\mathbf{f}_t(\mathbf{a}) = \left(\mathbf{\Phi}^T_{\mathbf{q}}\boldsymbol{\lambda} + (\mathbf{M}\ddot{\mathbf{q}} - \mathbf{g})\right)_t = \mathbf{0}, \quad t = 1,\ldots,n \\
\mathbf{0} \le \mathbf{a} \le \mathbf{1} \\
\left(\dfrac{\mathbf{f}^T_{\mathrm{GH}}\boldsymbol{\tau}_{\mathrm{GH}_d}}{\mathbf{f}^T_{\mathrm{GH}}\mathbf{n}_{\mathrm{GH}}}\right)_t \le th_d, \quad d = 1,2,3,4 \text{ and } t = 1,\ldots,n \\
(\mathbf{f}^T_{\mathrm{ST}_p}\mathbf{n}_{\mathrm{ST}_p})_t > 0, \quad p = 1,2 \text{ and } t = 1,\ldots,n
\end{cases}
\end{aligned} \tag{6}$$

where V^i is the volume, F^i_{CE} is the contractile force, $PCSA^i$ is the cross-sectional area, and a^i_t is the activation at time step t for muscle i. The variables c_1, c_2, and c_3 are weighting factors [43]. The constraints that the optimal problem has to fulfill are: (a) the equations of motion at each time t, represented by $\mathbf{f}_t(\mathbf{a})$; (b) the side constraints for the muscle activations; (c) the stability of the glenohumeral joint, in which $\mathbf{f}_{\mathrm{GH}}$ represents the glenohumeral joint reaction force, $\mathbf{n}_{\mathrm{GH}}$ the normal to the glenoid plane and $\boldsymbol{\tau}_{\mathrm{GH}_d}$ the unitary vector in the glenoid plane describing the direction d with threshold th_d; and, (d) the stability of the scapulothoracic joint, for which the reaction force is denoted by $\mathbf{f}_{\mathrm{ST}}$ and the directional vector of compression between the scapula and thorax by $\mathbf{n}_{\mathrm{ST}}$. This last constraint must be satisfied for each p point of the 2 defining points of the ST joint.

The optimal problem has $n \times (m + k)$ design variables, this is, for each of the n time steps there are m muscle activations and k Lagrange multipliers associated to the kinematic constraints, from which the joint reaction forces are found. The addition of $n \times k$ variables to the total number of design variables results from the fact that the joint reaction forces change with the muscle forces and thus the Lagrange multipliers associated with the kinematic joints must also be included in the optimization problem. As for the constraints, the problem has $n_{\mathrm{d.o.f.}} \times n$ of equality constraints, $6 \times n$ of inequality constraints and $n \times m$ of bound constraints. In this model, the total number of degrees of freedom $n_{\mathrm{d.o.f.}} = 42$, $m = 37$, and $k = 33$. The number of inequality constraints for each time step results from the 6 stability constraints defined for the GH and ST joints, i.e., 4 constraints regarding the glenohumeral contact force and 1 constraint regarding the scapulothoracic contact force of each of the 2 points defining the ST joint. Clearly, the size of the optimization problem is proportional to the number of time frames in the analysis which constitutes a limitation to the level of the analysis discretization that can be realistically used.

The biomechanical model is implemented in Matlab®. The Matlab function *fmincon* with the interior-point algorithm is used to solve the optimal problem. The objective function and constraint gradients are provided analytically to improve its performance while the Hessian is calculated by a limited-memory quasi-Newton approximation. Sampling the motion at constant-time intervals of 50 ms leads to a problem characterized by 9870 design variables, 5781 equality constraints, 846 inequality constraints, and 9870 lower bound and upper bound constraints.

3 Results

The optimal problem described by (6) is solved for several different initial sets of design variables. Given that the optimization algorithm used always finds local minima, this initialization procedure improves the chances that the best minimum found is in fact a global

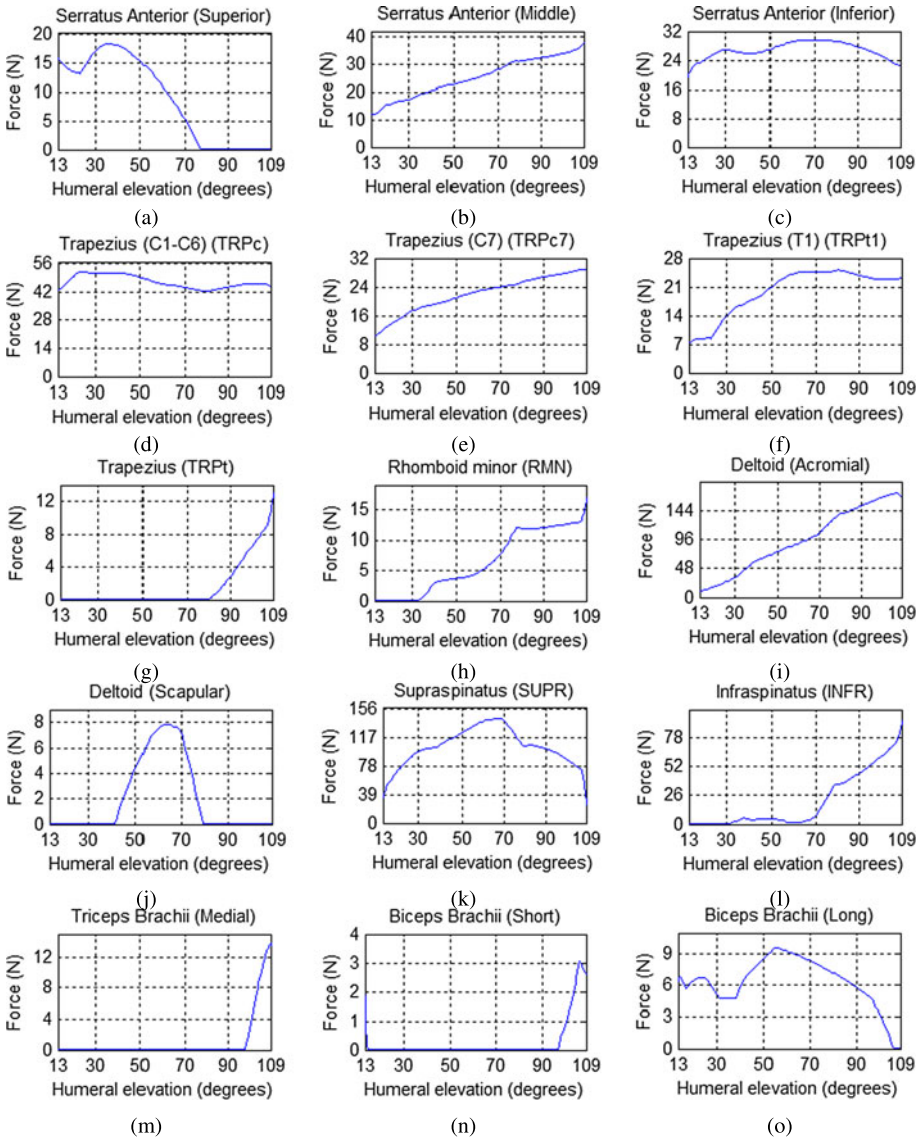


Fig. 6 Muscle forces, displayed in Newton, for the muscles crossing the SC, AC, and GH joints during arm abduction from 13° to 109°

minimum. The results, deemed as the best among the different trials, are summarized in Figs. 6 through 8. For the sake of conciseness, the issues concerning with the convergence of the optimization methods are not discussed here.

3.1 Muscle forces

Figures 6a to 6o present the contractile element forces predicted by the model for the muscles crossing the SC, AC, and GH joints during the abduction motion, where the arm gradu-

ally increases from 13° to 109° of elevation. The muscles generating less than 1 N of active force are hardly active and thus are not shown. In this range of motion, the contributions of the passive elements are also negligible and not shown in the figures.

The serratus anterior muscle is the prime mover in scapular protraction and upward rotation [45] being divided into three bundles in the model. The superior bundle, with the force shown in Fig. 6a, is active until 75° of humeral elevation while the middle bundle, with the force shown in Fig. 6b, and the inferior bundle, whose force is depicted by Fig. 6c, are active throughout the complete motion. The superior and middle fibers, due to their more horizontal line of pull are particularly responsible for the protraction of the scapula, while the lower fibers are effective in rotating the scapula upwardly by pulling it on the lower part. Moreover, the overall activity of the serratus anterior agrees with its major function of stabilizing the scapula against the rib cage [1].

The trapezius muscle is functionally divided into 3 parts: upper, middle, and lower parts [1]. The upper part, modeled by the TRPc bundle whose force is depicted by Fig. 6d, reaches its maximum activity at about 30°. Despite the difference in magnitude, its activity pattern is similar to that of the inferior fibers of the serratus anterior which fits well with their common function in the upward rotation of the scapula. The activity of the middle part, TRPc7 and TRPt1 bundles shown in Figs. 6e and 6f, increases with arm elevation. The action of the lower part, TRPt, seen in Fig. 6g, is important when acting together with the upper part to rotate the scapula and turn the glenoid fossa upward during abduction of the arm [46].

The rhomboid minor muscle, with the force shown in Fig. 6h, together with the serratus anterior muscle, contributes to the stabilization of the scapula over the thorax [45]. The model predicts its activity after 30° of amplitude.

The medial part of the deltoid, represented by DLTa in Fig. 6i, has the largest PCSA of the shoulder mechanism. Given its effectiveness in abducting the arm, it is expected to produce a large amount of force, especially between 30°–120° of amplitude [45]. The model proposed here supports this statement as it predicts the DLTa to be the most active muscle during abduction, reaching a maximum force of 173 N at the end of the analyzed motion. In [45], the anterior and posterior parts of the deltoid are said to act as stabilizers for the middle part during abduction yet the model only predicts some activity of the posterior part (DLTs), shown in Fig. 6j. The lack of activity of the clavicular part of the deltoid can be explained by the lack of ligament forces in the equations of motion. In [7], Van der Helm concluded that the preclusion of the conoid ligament from force transmission led to the inactivity of the clavicular part of the deltoid and the trapezius. Note that the sternoclavicular, costoclavicular, conoid, and trapezoid ligaments are modeled based on data published in [47]; however, their contribution is not taken into account in the equations of motion due to the lack of information regarding their rest length.

The supraspinatus muscle, with the force depicted by Fig. 6k, increases activity until 70°. During this period, it is in fact the most active muscle which reinforces not only its responsibility of initiating the abduction motion [46], but also assisting it and stabilizing, along with the other rotator cuff muscles, the glenohumeral joint [1, 45]. Among the remaining rotator cuff muscles, only the infraspinatus muscle, shown in Fig. 6l, displays noteworthy activity, especially after 70° of arm amplitude which coincides with the decrease in force of the supraspinatus muscle.

The triceps brachii muscle does not contribute to the arm abduction [45]. Nevertheless, the model predicts some activity for the medial head after 100° of amplitude, as seen in Fig. 6m. While recording motion at the LBL, the elbow is not restrained so the forearm must have moved, even if by a small amount, with respect to the arm. Note that the EMG detected some activity at the end of motion as well.

The biceps brachii muscle is modeled by two bundles according to its two heads, the short and long heads. The short head of the biceps displays some activity at the end of abduction, as seen in Fig. 6n, while the long head is active almost always, as seen in Fig. 6o. Vizniak relates the long head of the biceps with arm abduction [45]. Additionally, Terry and Chopp highlight the function of the long head of the biceps as a humeral head depressor, contributing to the stability of the shoulder joint [48].

3.2 EMG measured activity

As mentioned, the electromyographic activity is used here as a qualitative indicator of the muscle activity that can support the simulation results. The rms of the EMG signal recorded for 9 superficial muscles of the upper limb, namely the Serratus anterior, Trapezius p. descendenz, Trapezius p. transversus, Deltoideus p. clavicularis, Deltoideus p. acromialis, Deltoideus p. scapularis, Biceps brachii, Pectoralis Major, and Triceps brachii, is presented from Fig. 7a to Fig. 7i, respectively. To illustrate the period of muscle activity, a gray rectangle is drawn over the signal in all figures.

According to the EMG, the serratus anterior muscle is almost always active. Given the electrode position, the signal is more related to the medial part of the serratus anterior in the current model and thus confirms the model prediction. The trapezius *p. descendenz* is more related to the TRPc, TRPc7, and TRPt1 bundles. The EMG shows activity during all abduction, as the model predicted. The trapezius p. transversus, more related to the TRPt bundle, also resembles the activity pattern predicted by the model.

The acromial part of the deltoid is always active, as expected, and corroborates the results obtained. The scapular and clavicular parts of the deltoid, however, do not agree with the model. In the model the lack of activity of the clavicular part may be due to the preclusion of ligaments in the equations of motion; the scapular part, as it should play a similar role in the arm abduction as the clavicular part, may also be affected by it. It should be noted that the acromial, clavicular, and scapular electrodes are very close to each other, which may lead to some cross-talk between them. Furthermore, the electrode from the acromial part, positioned between the electrodes from the clavicular and scapular parts, exhibits the highest amplitude of all signals.

The biceps brachii signal agrees with the model prediction for the short head of the biceps. In any case, the recorded signal is not the most accurate one as the arm is not positioned in the best way to acquire the muscle EMG signal. Zipp [49] recommends the forearm to be supinated however to follow the ISB instructions, the forearm is completely pronated during the arm abduction.

Despite the small difference in arm amplitude at the beginning of muscle activity of the triceps brachii signal, it is in agreement with the model. The pectoralis major signal shows no activity as the model predicted.

3.3 Joint reaction forces

The SC, AC, ST, HU, and RU joint reaction forces are rather small, being almost always less than 10% of body weight, as observed in Figs. 8a, 8b, 8c, 8e, and 8f, respectively. The GH joint exhibits the highest reaction force, as seen in Fig. 8d, which is 32% of body weight at around 100° of arm elevation.

The results for the SC and AC joints are in close agreement with the Newcastle model [4]. Comparing to the Van der Helm's model [7], only the SC joint presents some similarity. Note that the Newcastle model includes the conoid and trapezoid ligaments and the Van

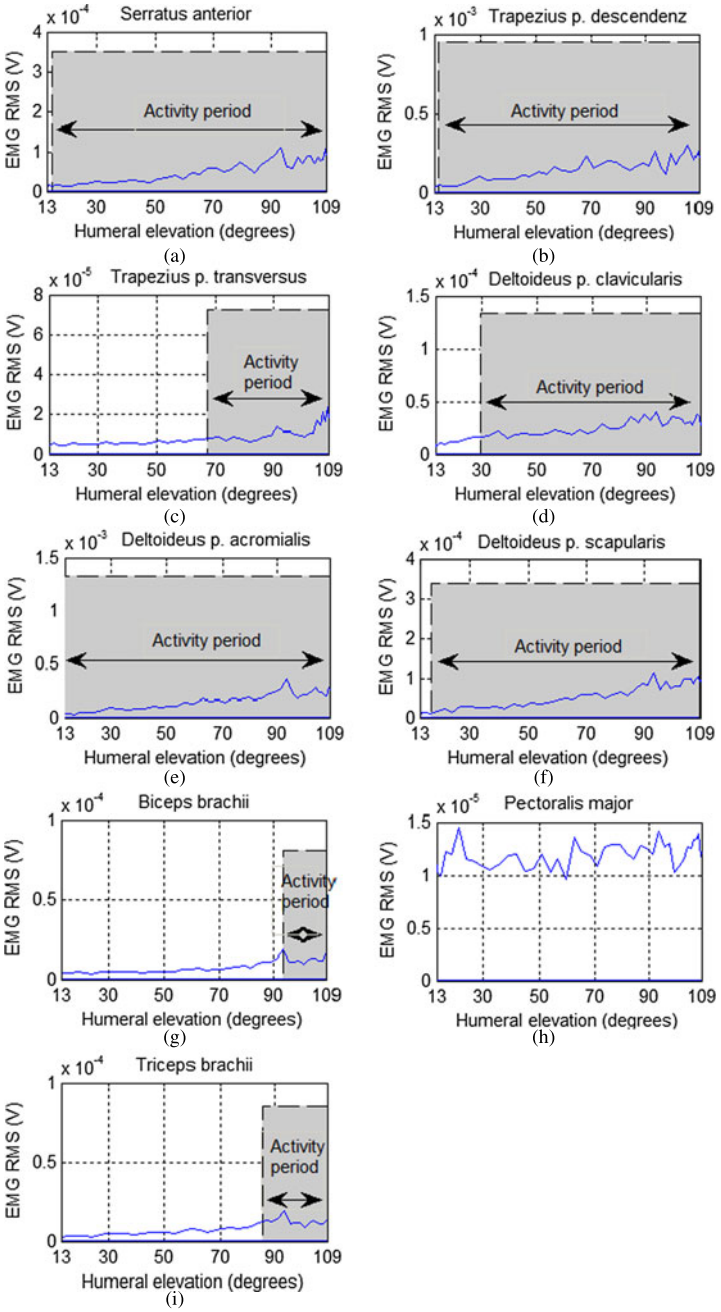


Fig. 7 RMS of the EMG signals and muscle on/off patterns

der Helm's model only includes the conoid ligament, so some differences are expected due to their contribution. Moreover, the results depend upon the movement which is certainly different for each model; besides being very difficult to reproduce the exact same movement

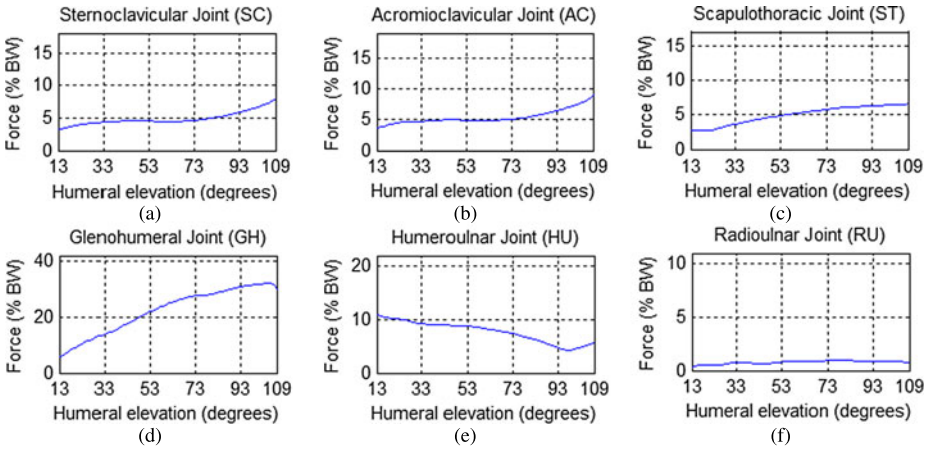


Fig. 8 Joint reaction forces estimated by the inverse dynamics analysis

more than once, different methods were used to define it. Van der Helm recorded static positions at different arm elevations; the Newcastle model uses the motion of the clavicle and scapula given by regression equations while in the current study motion is recorded using video imaging. These reasons also justify the differences regarding the ST joint reaction force magnitude. The model only reaches a maximum of 7% of body weight while in [4] and in [7] 15% of body weight was reached.

The GH joint reaction force matches closely the results from Van der Helm's model and the Newcastle model though with different magnitudes. The current model reached a maximum of 32% while the Van der Helm's model reaches about 50% and the Newcastle model reaches about 80% of the body weight.

4 Summary and conclusions

This work presents a biomechanical model of the upper limb developed and implemented through multibody formulations. The skeletal and musculoskeletal data needed to define the model was first presented by Garner and Pandy [16, 25] but transformed here to a new data set that uses new reference frames that follow the recommendations of the International Society of Biomechanics. The model is driven through an acquired motion by muscle actuators that drive the degrees-of-freedom of the joints crossed by the muscles. The motion was acquired using video imaging and a novel methodology was adopted to correct the rotations of the scapula [32]. The muscle force sharing problem is addressed through optimization tools. Instead of the standard static optimization, a global static optimization method is used in which the muscle activation and force history for all muscles is calculated during the complete movement. The results show that the muscle forces and joint reaction forces predicted are within the values reported in the literature [4, 7]. Furthermore, the activation patterns are qualitatively consistent with the known muscle functions [1, 45]. Comparing the results to the EMG data, only the clavicular and scapular parts of the deltoid do not match the expected values within reasonable limits.

As a future work, the model must take into account the ligaments contribution to the shoulder biomechanics. Clearly, the ligaments play an important role in maintaining shoul-

der stability and their inclusion into the model will yield more accurate muscle forces predictions [50]. Another objective is to simulate different pathological conditions, especially after a reverse shoulder arthroplasty (RSA). The RSA has gained popularity over the last few years due to their satisfactory functional results in patients with cuff-tear arthropathy and other difficult reconstructive shoulder problems [51]. However, the reverse prosthesis changes the natural biomechanics of the shoulder and understanding these changes is critical to attain the best possible outcomes [52]. The model proposed in this work provides a basis for the biomechanical model of the upper limb with shoulder prosthesis.

Acknowledgements This work was supported by the Foundation for Science and Technology through the project PTDC/SAU-BEB/103408/2008 and a Ph.D. scholarship—SFRH/BD/46311/2008. The authors are grateful to Daniel Simões Lopes, Paulo Luzio de Melo, and Nelson Ribeiro for their assistance at the LBL.

Appendix

The skeletal and musculoskeletal data described in the local reference frames of each body, whose definition is in agreement with the ISB, is presented in Tables 5 and 6. Table 5 presents the obstacle-set parameters used to model each muscle bundle in the model. The muscle path is always defined by an origin and an insertion site and, in between, any number of via points and obstacles may exist. Under the obstacle column, the single sphere, single cylinder, double cylinder, and sphere-capped cylinder are named sphere, single cylind., double cylinder, and stub, respectively. All coordinates presented are relative to the local reference frame described in the frame column. A more detailed explanation is given in [24, 25].

Table 5 Obstacle-set parameters used to model the muscle path of 21 muscles of the upper limb

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Subclavius (SBCL)	Origin		Thorax	−0.020580	0.005900	0.055880	
	Insert		Clavicle	−0.009532	−0.007983	0.014653	
Serratus anterior	Origin		Thorax	−0.056610	0.013200	0.085980	
(superior) (SRAs)	Single	Center	Thorax	−0.084900	0.018000	0.045370	−0.07888
	cylind.	x-Axis		0.134578	0.737302	0.662023	
		y-Axis		0.975759	−0.214962	0.041050	
		z-Axis		0.172576	0.640451	−0.748358	
	Insert		Scapula	−0.051469	0.039337	−0.007516	
Serratus anterior	Origin		Thorax	−0.012790	−0.073270	0.114810	
(middle) (SRAm)	Single	Center	Thorax	−0.071350	−0.058490	0.050190	−0.15582
	cylind.	x-Axis		−0.003986	0.207826	0.978158	
		y-Axis		0.981817	−0.184832	0.043272	
		z-Axis		0.189788	0.960544	−0.203310	
	Insert		Scapula	−0.077444	−0.037495	0.003240	

Table 5 (*Continued*)

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Serratus anterior	Origin		Thorax	0.012170	−0.188490	0.133240	
(inferior) (SRAi)	Single	Center	Thorax	−0.060190	−0.135350	0.031140	−0.22316
	cylind.	x-Axis		−0.279102	0.081165	0.956825	
		y-Axis		0.957817	−0.047525	0.283423	
		z-Axis		0.068477	0.995567	−0.064477	
	Insert		Scapula	−0.084064	−0.092847	−0.000958	
Trapezius (C1–C6)	Origin		Thorax	−0.085240	0.142730	0.001710	
(TRPc)	Single	Center	Thorax	−0.069970	0.090340	0.039220	0.0441
	cylind.	x-Axis		0.141858	0.245031	0.959081	
		y-Axis		0.982514	0.083181	−0.166575	
		z-Axis		−0.120593	0.965940	−0.228946	
	Insert		Scapula	−0.016578	0.001596	0.045698	
Trapezius (C7)	Origin		Thorax	−0.110510	0.071900	0.003950	
(TRPc7)	Single	Center	Thorax	−0.104530	0.039010	0.045460	0.0946
	cylind.	x-Axis		−0.007211	0.118960	0.992873	
		y-Axis		0.589782	−0.801310	0.100292	
		z-Axis		0.807530	0.586302	−0.064382	
	Insert		Scapula	0.053271	0.032469	0.011411	
Trapezius (T1)	Origin		Thorax	−0.124170	0.052180	0.000000	
(TRPt1)	Single	Center	Thorax	−0.115930	0.024740	0.048430	0.10306
	cylind.	x-Axis		0.010970	0.009010	0.999899	
		y-Axis		0.618811	−0.785540	0.000289	
		z-Axis		0.785464	0.618745	−0.014193	
	Insert		Scapula	−0.008219	0.034764	0.025560	
Trapezius (T2–T7)	Origin		Thorax	−0.172470	−0.109820	0.002900	
(TRPt)	Single	Center	Thorax	−0.106800	−0.069640	0.043890	0.14544
	cylind.	x-Axis		0.022615	0.014307	0.999642	
		y-Axis		0.993127	−0.115173	−0.020819	
		z-Axis		0.114834	0.993242	−0.016813	
	Insert		Scapula	−0.073950	−0.007748	0.010203	
Lev. Scap. 4 (LVS)	Origin		Thorax	−0.042510	0.147510	0.029630	
	Insert		Scapula	−0.053307	0.046731	−0.004321	
Rhomboid minor	Origin		Thorax	−0.094390	0.085340	0.001590	
(RMN)	Single	Center	Thorax	−0.066790	0.018740	0.042130	0.14166
	cylind.	x-Axis		0.042086	−0.076805	0.996157	
		y-Axis		0.868613	−0.489863	−0.074467	
		z-Axis		0.493700	0.868410	0.046097	
	Insert		Scapula	−0.067005	0.026075	0.004508	

Table 5 (*Continued*)

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Rhomboid major	Origin		Thorax	−0.125500	0.047970	0.000520	
(T1–T2) (RMJt2)	Single	Center	Thorax	−0.066790	0.018740	0.042130	0.15166
	cylind.	x-Axis		0.042086	−0.076805	0.996157	
		y-Axis		0.868613	−0.489863	−0.074467	
		z-Axis		0.493700	0.868410	0.046097	
	Insert		Scapula	−0.074474	−0.009819	0.007245	
Rhomboid major	Origin		Thorax	−0.149150	−0.007750	0.002490	
(T3–T4) (RMJt3)	Single	Center	Thorax	−0.092460	−0.051990	0.052130	0.1412
	cylind.	x-Axis		0.014501	0.090823	0.995761	
		y-Axis		0.955174	−0.295758	0.013066	
		z-Axis		0.295691	0.950936	−0.091041	
	Insert		Scapula	−0.085324	−0.064976	0.004605	
Pec. minor (PMN)	Origin		Thorax	−0.007640	−0.079220	0.108260	
	Insert		Scapula	0.035604	0.015424	−0.035426	
Pectoralis major	Origin		Clavicle	0.007808	−0.005234	−0.032057	
(clavicular) (PMJc)	Single	Center	Thorax	−0.075200	−0.019050	0.120450	−0.140
	cylind.	x-Axis		−0.549153	−0.453903	0.701715	
		y-Axis		0.690614	0.226361	0.686887	
		z-Axis		−0.470621	0.861819	0.189165	
	Insert		Humerus	0.005240	0.027797	0.016583	
Pectoralis major	Origin		Thorax	0.021920	−0.060150	0.028030	
(sternal) (PMJs)	Single	Center	Thorax	−0.087370	−0.073240	0.064490	−0.209
	cylind.	x-Axis		−0.092383	0.051844	0.994373	
		y-Axis		0.980988	−0.166429	0.099817	
		z-Axis		0.170667	0.984690	−0.035483	
	Insert		Humerus	0.007461	0.065624	0.016440	
Pectoralis major	Origin		Thorax	0.050910	−0.158580	0.034450	
(ribs) (PMJr)	Single	Center	Thorax	−0.071850	−0.065730	0.046280	−0.20286
	cylind.	x-Axis		−0.168136	0.209186	0.963313	
		y-Axis		0.984890	−0.005483	0.173093	
		z-Axis		0.041490	0.977860	−0.205103	
	Insert		Humerus	0.011364	0.102118	0.018144	
Latissimus dorsi	Origin		Thorax	−0.165150	−0.187070	0.001540	
(thoracic) (LTDt)	Double	Center	Thorax	−0.099280	−0.152060	0.056670	0.14888
	cylind.	x-Axis		0.982399	0.041482	0.182131	
		y-Axis		0.184377	0.371728	0.909848	
		z-Axis		−0.029961	0.927415	−0.372833	
		Center	Scapula	0.036207	−0.020636	0.001606	−0.04182
		x-Axis		0.586376	−0.479356	0.652979	
		y-Axis		0.316893	−0.606115	−0.729522	
		z-Axis		0.745481	0.634700	−0.203506	
	Insert		Humerus	0.003710	0.064121	0.006400	

Table 5 (*Continued*)

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Latissimus dorsi	Origin		Thorax	−0.143550	−0.360530	0.004620	
(lumbar) (LTDl)	Double	Center	Thorax	−0.095030	−0.224480	0.056130	0.16408
	cylind.	x-Axis		−0.278497	−0.109078	0.954223	
		y-Axis		0.939619	0.174797	0.294216	
		z-Axis		−0.198888	0.978544	0.053811	
		Center	Humerus	−0.008200	0.079638	0.002769	−0.02594
		x-Axis		0.051772	0.200831	0.978257	
		y-Axis		0.998153	−0.041578	−0.044290	
		z-Axis		0.037389	0.978266	−0.203856	
	Insert		Humerus	0.005011	0.082901	0.008447	
Latissimus dorsi	Origin		Thorax	−0.143550	−0.360530	0.004620	
(lumbar) (LTDl)	Double	Center	Thorax	−0.095030	−0.224480	0.056130	0.16408
	cylind.	x-Axis		−0.278497	−0.109078	0.954223	
		y-Axis		0.939619	0.174797	0.294216	
		z-Axis		−0.198888	0.978544	0.053811	
		Center	Humerus	−0.008200	0.079638	0.002769	−0.02594
		x-Axis		0.051772	0.200831	0.978257	
		y-Axis		0.998153	−0.041578	−0.044290	
		z-Axis		0.037389	0.978266	−0.203856	
	Insert		Humerus	0.005011	0.082901	0.008447	
Latissimus dorsi	Origin		Thorax	−0.128040	−0.419760	0.049550	
(iliac) (LTDi)	Double	Center	Thorax	−0.073810	−0.313520	0.095010	0.11776
	cylind.	x-Axis		−0.222249	−0.037328	0.974275	
		y-Axis		0.959100	0.171302	0.225351	
		z-Axis		−0.175307	0.984511	−0.002270	
		Center	Humerus	−0.008200	0.079638	0.002769	−0.02594
		x-Axis		0.051772	0.200831	0.978257	
		y-Axis		0.998153	−0.041578	−0.044290	
		z-Axis		0.037389	0.978266	−0.203856	
	Insert		Humerus	0.008313	0.104369	0.006064	
Deltoid	Origin		Clavicle	−0.007478	−0.006228	0.039853	
(clavicular) (DLTc)	Stub	Center	Scapula	0.029353	−0.016590	−0.031059	0.08430
		x-Axis		0.780871	0.190168	−0.595041	
		y-Axis		0.265291	−0.963328	0.040274	
		z-Axis		−0.565560	−0.189310	−0.802685	
	Via		Humerus	0.011133	0.054716	0.020872	
	Insert		Humerus	0.001509	0.039205	0.021475	

Table 5 (*Continued*)

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Deltoid (acromial)	Origin		Scapula	0.075220	0.018783	0.019865	
(DLTa)	Stub	Center	Scapula	0.049576	−0.017267	−0.004009	0.09256
		x-Axis		0.796470	0.025343	0.604146	
		y-Axis		−0.045315	−0.993811	0.101428	
		z-Axis		0.602976	−0.108163	−0.790392	
	Via		Humerus	−0.006168	0.036697	0.034270	
	Insert		Humerus	−0.001924	0.009165	0.024335	
Deltoid (scapular)	Origin		Scapula	0.008981	0.025917	0.030405	
(DLTs)	Via		Scapula	0.018854	0.006910	0.059724	
	Stub	Center	Scapula	0.030453	−0.023741	−0.011439	−0.14288
		x-Axis		−0.007219	−0.097865	0.995173	
		y-Axis		−0.885573	0.462851	0.039092	
		z-Axis		−0.464442	−0.881018	−0.090008	
	Via		Humerus	−0.027709	0.057766	0.030595	
	Insert		Humerus	−0.008896	0.043186	0.024107	
Supra-spinatus	Origin		Scapula	−0.023597	0.032954	−0.001227	
(SUPR)	Via		Scapula	0.033797	0.025283	0.004086	
	Via		Humerus	−0.005172	0.161214	0.023195	
	Insert		Humerus	−0.003434	0.151238	0.029676	
Infra-spinatus	Origin		Scapula	−0.041348	−0.025199	0.015032	
(INFR)	Via		Scapula	0.026955	−0.013907	0.031414	
	Stub	Center	Scapula	−0.004536	0.142218	0.004976	0.05106
		x-Axis		−0.793738	0.325941	−0.513558	
		y-Axis		−0.483202	0.174944	0.857853	
		z-Axis		0.369453	0.929062	0.018636	
	Insert		Humerus	−0.008090	0.142485	0.030857	
Sub-scapularis	Origin		Scapula	−0.040439	−0.031019	0.004506	
(SBSC)	Via		Scapula	0.005170	−0.016085	−0.016511	
	Single	Center	Scapula	0.023192	0.012796	−0.008592	0.03082
	cylind.	x-Axis		0.444443	−0.111244	−0.888872	
		y-Axis		0.895001	0.013080	0.445871	
		z-Axis		−0.037974	−0.993709	0.105375	
	Insert		Humerus	0.016772	0.141293	−0.003379	
Teres minor	Origin		Scapula	−0.010403	−0.045772	0.018824	
(TMN)	Via		Scapula	0.005450	−0.034259	0.031051	
	Stub	Center	Humerus	−0.004471	0.141199	0.010807	0.04417
		x-Axis		−0.812242	−0.073322	−0.578693	
		y-Axis		−0.552557	−0.221267	0.803583	
		z-Axis		−0.186940	0.972463	0.139172	
	Insert		Humerus	−0.008878	0.114531	0.027503	

Table 5 (*Continued*)

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Teres major (TMJ)	Origin		Scapula	−0.063192	−0.081645	0.016139	
	Single	Center	Scapula	−0.037742	−0.078844	0.000739	0.04800
	cylind.	x-Axis		−0.617827	0.403449	0.674921	
		y-Axis		0.528484	−0.422504	0.736338	
		z-Axis		0.582231	0.811617	0.047819	
	Insert		Humerus	−0.004359	0.090834	0.000177	
Coraco-brachialis	Origin		Scapula	0.047180	0.004295	−0.033638	
(CRCB)	Single	Center	Scapula	0.013194	−0.030664	0.002898	0.03462
	cylind.	x-Axis		0.182214	−0.787608	−0.588620	
		y-Axis		0.422983	−0.477630	0.770035	
		z-Axis		−0.887627	−0.389288	0.246112	
	Via		Humerus	0.004393	0.039466	−0.003037	
	Insert		Humerus	−0.009315	−0.023481	0.005695	
Triceps brachii	Origin		Scapula	0.022362	−0.030890	0.012474	
(long) (TRClg)	Double	Center	Humerus	0.071248	0.005701	0.027168	0.24438
	cylind.	x-Axis		0.982548	0.152910	−0.105917	
		y-Axis		−0.185258	0.855620	−0.483315	
		z-Axis		0.016721	0.494502	0.869015	
		Center	Humerus	−0.002512	−0.154233	0.010295	0.058
		x-Axis		−0.597712	0.801133	0.030436	
		y-Axis		−0.790585	−0.595295	0.143522	
	Insert		Ulna	−0.027590	0.120980	−0.003652	
Triceps brachii	Origin		Humerus	−0.021626	−0.031157	0.003596	
(medial) (TRCm)	Via		Humerus	−0.015317	−0.080974	−0.014857	
	Single	Center	Humerus	−0.007386	−0.161108	0.024148	0.05008
	cylind.	x-Axis		−0.455914	0.885785	−0.086757	
		y-Axis		−0.874561	−0.463951	−0.141042	
		z-Axis		−0.165183	0.011571	0.986195	
	Insert		Ulna	−0.027590	0.120980	−0.003652	
Triceps brachii	Origin		Humerus	−0.018593	0.051375	0.011804	
(lateral) (TRClt)	Via		Humerus	−0.039143	−0.009284	0.029226	
	Single	Center	Humerus	−0.008121	−0.157218	0.012194	0.04558
	cylind.	x-Axis		−0.565142	0.812901	0.140736	
		y-Axis		−0.771786	−0.581215	0.257944	
		z-Axis		0.291480	0.037157	0.955855	
	Via		Ulna	−0.027740	0.116830	0.004928	
	Insert		Ulna	−0.027590	0.120980	−0.003652	

Table 5 (*Continued*)

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Biceps brachii	Origin		Scapula	0.051317	0.008236	−0.026231	
(short) (BICs)	Double	Center	Scapula	0.046295	−0.013204	−0.003694	0.04854
	cylind.	x-Axis		−0.498504	0.736735	0.456929	
		y-Axis		0.468255	0.672356	−0.537345	
		z-Axis		−0.729622	−0.071844	−0.680066	
		Center	Humerus	−0.031815	−0.137995	0.028451	0.1029
		x-Axis		0.098752	0.857697	0.504583	
		y-Axis		0.944683	0.078567	−0.318433	
		z-Axis		−0.312763	0.508117	−0.802494	
	Insert		Radius	0.011267	0.062358	−0.006852	
Biceps brachii	Origin		Scapula	0.044350	0.007356	−0.005102	
(long) (BICl)	Sphere	Center	Humerus	−0.005207	0.137592	0.007898	0.04850
	Via		Humerus	0.018847	0.136860	0.0130018	
	Single	Center	Humerus	−0.031815	−0.137995	0.028451	0.1029
	cylind.	x-Axis		0.098752	0.857697	0.504583	
		y-Axis		0.944683	0.078567	−0.318433	
		z-Axis		−0.312763	0.508117	−0.802494	
	Insert		Radius	0.011267	0.062358	−0.006852	
Brachialis (BRA)	Origin		Humerus	−0.004796	−0.078525	0.013732	
	Via		Humerus	0.006898	−0.102462	0.014796	
	Single	Center	Humerus	−0.003257	−0.150069	0.070357	0.07046
	cylind.	x-Axis		0.331567	0.941483	−0.060598	
		y-Axis		0.910935	−0.336199	−0.239096	
		z-Axis		−0.245478	0.024076	−0.969103	
	Insert		Ulna	−0.006370	0.079420	−0.001182	
Brachio-radialis	Origin		Humerus	−0.015876	−0.081455	0.024688	
(BRD)	Via		Humerus	−0.010508	−0.092090	0.031797	
	Double	Center	Humerus	−0.019750	−0.157746	0.013999	−0.06456
	cylind.	x-Axis		−0.568861	0.815252	0.108450	
		y-Axis		−0.704815	−0.551205	0.446553	
		z-Axis		0.423832	0.177589	0.888161	
		Center	Radius	−0.004560	−0.042135	−0.010147	0.01434
		x-Axis		−0.999165	0.037250	−0.016761	
		y-Axis		−0.019037	−0.061680	0.997915	
		z-Axis		0.036140	0.997400	0.062336	
	Via		Radius	−0.004973	−0.079347	−0.004727	
	Insert		Radius	0.001461	−0.150209	−0.004018	

Table 5 (*Continued*)

Muscle	Obstacle		Frame	x (m)	y (m)	z (m)	Diam. (m)
Supinator (SUP)	Origin		Ulna	−0.023100	0.062590	0.017318	
	Single	Center	Radius	−0.006050	0.012568	−0.016263	0.02572
	cylind.	x-Axis		0.256671	−0.040254	0.965660	
		y-Axis		0.966192	−0.014477	−0.257416	
		z-Axis		0.024343	0.999084	0.035177	
	Via		Radius	−0.005898	0.025620	0.006089	
	Insert		Radius	0.002530	0.018862	−0.005433	
Pronator teres	Origin		Humerus	−0.018975	−0.144200	−0.004172	
(PRNT)	Via		Humerus	−0.007357	−0.153144	−0.013845	
	Single	Center	Humerus	−0.003163	−0.169181	0.013604	0.03516
	cylind.	x-Axis		0.289872	0.866061	−0.407324	
		y-Axis		0.632725	−0.492735	−0.597387	
		z-Axis		−0.718077	−0.084558	−0.690808	
	Insert		Radius	0.000561	−0.009491	−0.002762	

Table 6 Bony landmarks coordinates defined in the local reference frames of each body. The bony landmarks HUC and PHUC represent the HU joint center and a point on the HU joint axis, respectively. The same meaning have RUC and PRUC but for the RU joint. The remaining bony landmarks are explained in Table 4

Frame	Bony landmark	x (m)	y (m)	z (m)
Thorax	IJ	0.000000	0.000000	0.000000
	SC	−0.014380	−0.003710	0.032540
Clavicle	SC	−0.001264	−0.017004	−0.073247
	AC	−0.001265	−0.017003	0.080923
Scapula	AC	0.059495	0.032658	0.000571
	TS	−0.063009	0.032659	0.000575
	TS projected on the rib cage	−0.057789	0.020745	−0.025387
	AI	−0.088414	−0.088411	0.000585
	AI projected on the rib cage	−0.083834	−0.085858	−0.020842
	GH	0.059398	−0.010938	0.005177
Humerus	GH	−0.005207	0.137592	0.007898
	HUC	−0.005217	−0.166743	0.007900
	PHUC	0.016706	−0.163002	0.105396
Ulna	EJC	−0.002700	0.110990	−0.003702
	PEJC	−0.002700	0.089398	0.093939
	US	−0.002700	−0.146120	−0.003702
	RUC	−0.002700	0.110990	0.020118
	PRUC	−0.002700	0.210564	0.029344
Radius	RUC	−0.002700	0.111000	−0.003700
	PRUC	−0.002854	0.209900	0.011091
	RS	0.001480	−0.163402	−0.006672

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